18

Primality testing

We want to know whether a given integer is prime or not. Certainly we can find out by factoring it. Can you think of any other way? Well, there is, and the major discovery in this area is that primality testing is much easier than factoring, at least to current knowledge. One can test integers with many thousands of digits, but factoring numbers with only 300 digits is in general not feasible.

In this chapter, we provide an efficient probabilistic algorithm to test primality; factorization is the subject of the next chapter. As an easy application, we can also find large prime numbers, as they are required in some modular algorithms and in modern cryptography. We conclude with brief discussions of other primality testing algorithms. The long-standing quest for a deterministic polynomial-time primality test, stated as a Research Problem in the first two editions of this book, was resolved by Agrawal, Kayal & Saxena (2004).

For numbers of a special form, such as the **Mersenne numbers** $M_n = 2^n - 1$, particularly efficient methods have been known since the 19th century. Indeed, throughout history the largest known prime has usually been a Mersenne prime. On 23 August 2008, a UCLA computer managed by Edson Smith discovered $M_{43\,112\,609}$, the largest among 47 known Mersenne primes. It has 12 978 189 decimal digits. This current world record (at the time of writing) is an achievement of the Great Internet Mersenne Prime Search (GIMPS), based on software by George Woltman and Scott Kurowski. This record, together with another one, discovered two weeks later and slightly smaller, earned a US$ 100 000 award by the Electronic Frontier Foundation. GIMPS harnesses the spare power of over 20 000 computers all over the world, and performs about 720 billion calculations per second. This new paradigm of *internet computing* started in the area of integer factorization, and may solve in the future very large instances of such easily distributed problems.

18.1. Multiplicative order of integers

An integer $N \in \mathbb{N}_{\geq 2}$ is a **prime number** if $N \mid ab$ implies that $N \mid a$ or $N \mid b$, or, equivalently, if $ab = N$ implies that $a \in \{1, -1\}$ or $b \in \{1, -1\}$, for all $a, b \in \mathbb{Z}$

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(Section 25.2). The other integers $N \geq 2$ are **composite**. (The number 1 is neither prime nor composite, but a unit. In the ring $\mathbb{Z}$, -5 is a prime just as 5 is; see Section 3.1 for a discussion of **associates**.) The prime numbers form the "building blocks" for all integers, according to the following basic fact.

FUNDAMENTAL THEOREM OF ARITHMETIC. Every positive integer can be written as a product of positive prime factors. This is unique up to the order of the factors. In other words, $\mathbb{Z}$ is a Unique Factorization Domain.

The idea of primality was known to the Pythagoreans (around 500 BC), and Book IX of Euclid's *Elements* contains his famous proof that there are infinitely many primes (see page 26).

This chapter deals mainly with testing whether a given integer N is prime or not. Can we do better than to try division by all integers up to $\sqrt{N}$, a method already known to Eratosthenes in the third century BC? Yes, indeed, there are efficient algorithms that differentiate between prime and composite numbers, and as a result the set of prime numbers is in $\mathcal{P}$. In this text, we only present a probabilistic algorithm. It shows that this set is in the complexity class $\mathcal{ZPP}$ (see Section 25.8), and is eminently practical.

We recall the following facts. $\mathbb{Z}_N^\times = \{a \bmod N \in \mathbb{Z}_N : \gcd(a,N) = 1\}$ is the multiplicative group of units in $\mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z}$. Remember that a unit in a ring is an element that has an inverse in the ring. The elements of $\mathbb{Z}_N^\times$ form a multiplicative group of cardinality $\varphi(N) = \#\mathbb{Z}_N^\times$; φ is **Euler's totient function**. If the prime factorization of N is $N = p_1^{e_1} \cdots p_r^{e_r}$, where $p_1, \ldots, p_r$ are distinct positive primes and $e_1, \ldots, e_r$ are positive integers, then the Chinese Remainder Theorem 5.3 says that $\mathbb{Z}_N \cong \mathbb{Z}_{p_1^{e_1}} \times \cdots \times \mathbb{Z}_{p_r^{e_r}}$ (a ring isomorphism), and that $\mathbb{Z}_N^\times \cong \mathbb{Z}_{p_1^{e_1}}^\times \times \cdots \times \mathbb{Z}_{p_r^{e_r}}^\times$ (a group isomorphism). If N is prime, then $\mathbb{Z}_N$ is a field, and $\mathbb{Z}_N^\times$ is a group of order $\varphi(N) = N - 1$. If $N = p^e$ is a prime power, then $\varphi(N) = p^{e-1}(p-1)$, and in general, $\varphi(N) = p_1^{e_1-1} \cdot (p_1 - 1) \cdots p_r^{e_r-1} \cdot (p_r - 1)$, by Corollary 5.6.

A central fact is **Fermat's little theorem** 4.9 which says that $a^{N-1} \equiv 1 \bmod N$ for a prime N and any $a \in \mathbb{Z}$ which is coprime to N. For coprime integers a, N we define the **order** $\operatorname{ord}_N(a)$ of a modulo N as the smallest integer $k \geq 1$ such that $a^k \equiv 1 \bmod N$. **Euler's theorem**, generalizing Fermat's, states that $a^{\varphi(N)} \equiv 1 \bmod N$, and is a consequence of Lagrange's theorem (Section 25.1). Besides these "upper bounds" on the order, we also need some "lower bounds".

LEMMA 18.1. *Let $N \in \mathbb{N}_{\geq 2}$.*

(i) If $a \in \mathbb{Z}$ is coprime to N and $k \in \mathbb{N}$ with $a^k \equiv 1 \bmod N$, then $\operatorname{ord}_N(a)$ divides k. In particular, $\operatorname{ord}_N(a)$ divides $\varphi(N)$.

(ii) If p is prime, $e \in \mathbb{N}_{\geq 2}$, $N = p^e$, and $a = 1 + p^{e-1}$, then $\operatorname{ord}_N(a) = p$.

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PROOF. (i) Let $e = \mathrm{ord}_N(a)$ and divide k by e with remainder: $k = qe + r$ with $0 \le r < e$. Then $a^r = a^{k-qe} = a^k \cdot (a^e)^{-q} \equiv 1 \bmod N$, and hence $r = 0$. By Euler's theorem, we have $a^{\varphi(N)} \equiv 1 \bmod N$, and (i) follows.

(ii) We have $a^p \equiv \sum_{0 \le i \le p} \binom{p}{i} p^{(e-1)i} \equiv 1 \bmod p^e$. By (i), $\mathrm{ord}_N(a)$ is either 1 or p, and since $a \not\equiv 1 \bmod N$, the claim follows. □

If p is a prime, then in the language of Section 8.2, $a \in \mathbb{Z}$ is a primitive kth root of unity in $\mathbb{F}_p$ if and only if $p \nmid a$ and $\mathrm{ord}_p(a) = k$.

18.2. The Fermat test

From now on, we assume that $N \in \mathbb{N}_{\ge 3}$ is odd; testing even numbers for primality is rather easy. We recall that $a \text{ rem } N \in \mathbb{N}$ is the remainder of $a \in \mathbb{Z}$ on division by N, with $0 \le a \text{ rem } N < N$. We often use it to represent $a \bmod N \in \mathbb{Z}_N$, but the two objects live in different domains. Fermat's little theorem furnishes the **Fermat test** for primality.

ALGORITHM 18.2 Fermat test.
Input: An odd integer $N \ge 5$.
Output: Either "composite" or "possibly prime".

1. choose $a \in \{2, \cdots, N-2\}$ uniformly at random
2. **call** the repeated squaring algorithm 4.8 to compute $b = a^{N-1} \text{ rem } N$
3. **if** $b \ne 1$ **then return** "composite" **else return** "possibly prime"

If a and N are not coprime, then neither are b and N, and the algorithm correctly returns "composite". So we may assume that $\gcd(a, N) = 1$. Then by Fermat's little theorem, the answer is correct if the test replies "composite". If it replies "possibly prime", it may be right or it may be wrong. We have to understand when and why an error may occur. To this end, we consider the subgroup of

$$L_N = \{u \in \mathbb{Z}_N^\times : u^{N-1} = 1\}$$

of $\mathbb{Z}_N^\times$. Clearly L_N is a group, and Fermat's little theorem says that $L_N = \mathbb{Z}_N^\times$ if N is prime. If $L_N \ne \mathbb{Z}_N^\times$, then in fact $\#L_N \le \frac{1}{2}\#\mathbb{Z}_N^\times$, since the size of a finite group is an integer multiple of the size of any of its subgroups, by Lagrange's theorem (Section 25.1). If the a chosen in step 1, taken modulo N, happens to be in $\mathbb{Z}_N^\times \setminus L_N$, then the test will answer "composite". Such an a, and also its residue class $a \bmod N$, is called a **Fermat witness** to the compositeness of N. Similarly, if $a \bmod N \in L_N$, then a (and also $a \bmod N$) is a **Fermat liar** for N.

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If we know any Fermat witness, we are guaranteed that N is composite, although we still do not necessarily know a factor of N. The required computation is fairly easy to do. We have come across a famous historical example on page 513: the first five **Fermat numbers** $F_n = 2^{2^n}+1$ are prime, for $n = 0,1,2,3,4$. Pierre Fermat conjectured in August 1640 that all these numbers are prime, while he communicated his famous "little" theorem in a letter dated 18 October 1640. However, $3^{2^{32}} \equiv 1\,461\,798\,105 \not\equiv 1 \bmod 2^{32}+1$; therefore, by Fermat's little theorem, Fermat's conjecture is false. The required computation amounts to 32 multiplications modulo the 10-digit number $2^{32}+1$ (it takes a few milliseconds on a workstation). We saw in Section 4.3 how Euler had found the factor 641 of F_5. It turns out that for "most" composite N, most a's are Fermat witnesses. The impressive recent progress in factorization of Fermat numbers is presented in Section 19.1.

It would be nice if this test were enough. However, it clearly fails if N is composite and $L_N = \mathbb{Z}_N^\times$. Are there such numbers? Indeed there are, and we define the **Carmichael numbers** to be precisely those composite integers N for which $L_N = \mathbb{Z}_N^\times$. In other words, they are the composite numbers without any Fermat witnesses.

For the cost analysis, we recall the multiplication time $\mathsf{M}(n)$ for multiplying two integers of length n (Definition 8.26). We can use $\mathsf{M}(n) \in O(n\log n\operatorname{loglog} n)$, but the reader may also think of classical multiplication with $\mathsf{M}(n) \in O(n^2)$ (see inside back cover).

THEOREM 18.3.
If N is prime, then the Fermat test 18.2 returns "possibly prime". If N is composite and not a Carmichael number, then it returns "composite" with probability at least $1/2$. The algorithm uses $O(\log N \cdot \mathsf{M}(\log N))$ word operations.

PROOF. If $\gcd(a,N) > 1$, then also $\gcd(b,N) > 1$, and the test returns "composite", so that we only need to consider the cases where a and N are coprime. If N is composite and not Carmichael, then $\#L_N \le \varphi(N)/2$, as noted above, so that at least half of the possible choices for a in step 1 (coprime to N) are Fermat witnesses. Repeated squaring in step 2 takes $O(\log N)$ multiplications modulo N or $O(\log N\,\mathsf{M}(\log N))$ word operations, and the bound on the running time follows. □

If N is a Carmichael number, then the Fermat test returns either "possibly prime" or "composite". The latter happens only when $\gcd(a,N) > 1$.

18.3. The strong pseudoprimality test

What to do with the challenge posed by the Carmichael numbers? Pomerance (1990) pleads: *Using the Fermat congruence is so simple, that it seems a shame to give up on it just because there are a few counter-examples!*

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We now resolve the shortcoming of the Fermat test in a drastic way: the new test not only distinguishes primes from Carmichael numbers, it actually factors these seemingly difficult numbers in random polynomial time. In general, factoring integers is much harder than testing them for primality, and so these numbers turn out to be quite harmless after all.

LEMMA 18.4. *Any Carmichael number is squarefree.*

PROOF. We take a prime number p and assume that it divides the Carmichael number N exactly $e \geq 2$ times. By the Chinese Remainder Theorem, there exists an $a \in \mathbb{Z}$ with $a \equiv 1 + p^{e-1} \bmod p^e$ and $a \equiv 1 \bmod N/p^e$. Then a has order p modulo p^e, by Lemma 18.1, and hence also modulo N. Since $a^{N-1} \equiv 1 \bmod N$, it follows that p divides $N-1$, by the same lemma. Since p also divides N, we have a contradiction, and the claim is proved. □

Exercise 18.9 shows that N is a Carmichael number if and only if N is squarefree and $p-1$ divides $N-1$ for any prime factor p of N, and that Carmichael numbers are odd and have at least three prime factors. The first three Carmichael numbers are: $561 = 3 \cdot 11 \cdot 17, 1105 = 5 \cdot 13 \cdot 17$, and $1729 = 7 \cdot 13 \cdot 19$.

The following algorithm refines the Fermat test, and there is no input on which it systematically fails. If it is run on a Carmichael number, then it probably not only recognizes it as composite but in addition finds a proper factor of it.

ALGORITHM 18.5 Strong pseudoprimality test.
Input: An odd integer $N \geq 3$.
Output: Either "composite", or "probably prime", or a proper factor of N.

1. choose $a \in \{2, \ldots, N-2\}$ uniformly at random

2. $d \longleftarrow \gcd(a, N)$
   if $d > 1$ **then return** d

3. write $N - 1 = 2^v m$ with $v, m \in \mathbb{N}$, $v \geq 1$, and m odd
   call the repeated squaring algorithm 4.8 to compute $b_0 = a^m \text{ rem } N$
   if $b_0 = 1$ **then return** "probably prime"

4. **for** $i = 1, \ldots, v$ **do** $b_i \longleftarrow b_{i-1}^2 \text{ rem } N$

5. **if** $b_v = 1$ **then** $j \longleftarrow \min\{0 \leq i < v : b_{i+1} = 1\}$ **else return** "composite"

6. $g \longleftarrow \gcd(b_j + 1, N)$
   if $g = 1$ or $g = N$ **then return** "probably prime" **else return** g

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THEOREM 18.6.
If N is prime, then Algorithm 18.5 returns "probably prime". If N is composite and not a Carmichael number, then the algorithm returns "composite" with probability at least $1/2$. If N is a Carmichael number, the algorithm returns a proper divisor of N with probability at least $1/2$. It uses $O(\log N \cdot \mathsf{M}(\log N))$ word operations.

PROOF. By induction, we have $b_i \equiv a^{2^i m} \bmod N$ for $0 \le i \le v$, and in particular $b_v \equiv a^{N-1} \bmod N$. If $b_{i-1} = 1$, then also $b_i = 1$, for any i. If N is composite and not Carmichael, then with probability at least $1/2$, a is a Fermat witness for N, $b_v \neq 1$, and the algorithm returns "composite" in step 5. We next assume that N is prime. Then $b_v = 1$. If $b_0 = 1$, then the algorithm correctly returns "probably prime" in step 3. Otherwise, we have $b_j \neq 1$ and $b_j^2 \equiv b_{j+1} = 1 \bmod N$ in step 6. By Lemma 25.4, the polynomial $x^2 - 1 \in \mathbb{Z}_N[x]$ has at most two zeroes. Hence the only square roots of 1 modulo N are 1 and -1, so that $b_j = N - 1$ and $g = N$, and the correct result is returned in step 6.

The last case to be considered is when N is a Carmichael number. We let P be the set of prime divisors of N. Since N is squarefree, we have $N = \prod_{p \in P} p$. We consider

$$I = \{i : 0 \le i \le v \text{ and } \forall u \in \mathbb{Z}_N^\times \; u^{2^i m} = 1\}.$$

Then $v \in I$, by the definition of Carmichael numbers, and $i + 1 \in I$ for any $i \in I$ with $i < v$. Since m is odd, we have $(-1)^m = -1 \neq 1$, and therefore $0 \notin I$. Hence there exists some $l < v$ such that $I = \{l + 1, l + 2, \ldots, v\}$. Now let

$$G = \{u \in \mathbb{Z}_N^\times : u^{2^l m} = \pm 1\} \subseteq \mathbb{Z}_N^\times.$$

This is a subgroup of $\mathbb{Z}_N^\times$, and we now show that $G \neq \mathbb{Z}_N^\times$. There exists some $p \in P$ and $b \in \mathbb{Z}$ coprime to p with $b^{2^l m} \not\equiv 1 \bmod p$, since otherwise we would have $l \in I$. We take some such p and b. The Chinese Remainder Theorem implies that there exists a $c \in \mathbb{Z}$ such that $c \equiv b \bmod p$ and $c \equiv 1 \bmod N/p$. Then $c \bmod N \in \mathbb{Z}_N^\times \setminus G$. Being a proper subgroup, G has at most $\#\mathbb{Z}_N^\times / 2 = \varphi(N)/2$ elements.

If a in step 1 is chosen so that $a \bmod N \in \mathbb{Z}_N^\times \setminus G$, then we claim that the algorithm will actually discover a proper divisor of N. The fact that $b_{l+1} \equiv a^{2^{l+1} m} \equiv 1 \bmod N$ implies that for all $p \in P$, also $b_{l+1} \equiv 1 \bmod p$. Again, the only square roots of 1 modulo p are 1 and -1, so that for each p, $a^{2^l m} \bmod p$ is either 1 or -1. Since $b_l \bmod N = a^{2^l m} \bmod N$ is neither 1 nor -1, both possibilities actually occur, we have $j = l$ in step 5, and

$$g = \gcd(b_l + 1, N) = \prod_{\substack{p \in P \\ a^{2^l m} \equiv -1 \bmod p}} p$$

is a proper divisor of N. The fact that $\#(\mathbb{Z}_N^\times \setminus G) \ge \varphi(N)/2$ implies the bound on the probability.

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Steps 2 and 6 take $O(\mathsf{M}(\log N)\operatorname{loglog} N)$ word operations, as discussed at the end of Section 11.1, the cost of steps 3 and 4 is $O(\log N)$ multiplications modulo N or $O(\log N\,\mathsf{M}(\log N))$ word operations, and the time estimate follows. □

If N is composite, then a number $a \in \{1,\dots,N-1\}$, coprime to N, such that Algorithm 18.5 outputs "composite" or a proper factor of N is called a **strong witness** for the compositeness of N, and if "probably prime" is returned, it is a **strong liar**. We also use these two notions for the corresponding residue class $a \bmod N$. Every Fermat witness is a strong witness.

In order to push the error probability below some given positive ε, we can run the test independently $\lceil \log_2 \varepsilon^{-1} \rceil$ times and return "probably prime" if this is the result of each run, and otherwise "composite" or a factor.

In the proof of Theorem 18.3 and in step 2 of Algorithm 18.5, we have dealt separately with the case where $\gcd(a,N) > 1$. This is highly unlikely to happen when N is reasonably large.

What does it mean when a primality test returns "probably prime" on input N? Is N then "probably prime"? Of course not; N is either prime or it is not. The "probably" only refers to the random choices we made within the algorithm. (This is not to be confused with the weaker notion of "average case analysis", where one assumes a probability distribution on the inputs.) If we have run the test 1001 times, say, then it means the following: if N is not prime, then an event has been witnessed whose probability is at most 2^{-1001}. If you fly in an airplane whose safety depends on the actual primality of such an "industrial-strength pseudoprime", then this fact should not worry you unduly, since other things are much more likely to fail ;-)

Exercise 18.6 shows that with a small modification, both the Fermat test and the strong pseudoprimality test find the prime factor p when $N = p^e$ is a prime power. Exercise 18.12 shows how to modify the algorithm to find all prime factors of a Carmichael number (whose factors are in general not Carmichael numbers).

18.4. Finding primes

For our modular algorithms and in cryptography, one needs prime numbers satisfying certain specifications. For example, we may be given an integer n and require an n-bit prime. In this section, we provide the tools for a variety of such tasks, and the corollaries at the end give cost estimates for all our prime-consuming algorithms. In most (but not all) cases, this cost is negligible when compared to the overall time taken by the algorithms.

The famous **prime number theorem** in analytic number theory says approximately how many primes there are in an initial segment of the natural numbers.

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It may be stated in two equivalent ways, using the two functions

$$\pi(x) = \#\{p \in \mathbb{N} : p \le x, p \text{ prime}\}, \qquad p_n = \text{the } n\text{th prime number},$$

where $x \in \mathbb{R}_{>0}$ and $n \in \mathbb{N}_{>0}$ (so that $p_1 = 2$). Its proof is beyond the scope of this text.

THEOREM 18.7 *Prime number theorem.*
Let ln *denote the logarithm in base* e*. Then we have approximately*

$$\pi(x) \approx \frac{x}{\ln x}, \quad p_n \approx n \ln n,$$

and more precisely

$$\begin{aligned} \frac{x}{\ln x}\left(1+\frac{1}{2\ln x}\right) &< \pi(x) < \frac{x}{\ln x}\left(1+\frac{3}{2\ln x}\right) && \text{if } x \ge 59, \\ n\left(\ln n + \ln\ln n - \frac{3}{2}\right) &< p_n < n\left(\ln n + \ln\ln n - \frac{1}{2}\right) && \text{if } n \ge 20. \end{aligned}$$

The first statement says that a random integer near x is prime with probability about $(\ln x)^{-1}$. If we choose random n-bit integers and test them for primality, we expect to find a prime with about $n \cdot \ln 2$ trials. Throughout this section, "ln" is the "natural" logarithm for the prime number theory, but we continue to use "log" in "O" estimates of running times, where the base is irrelevant.

In order to find a large prime p, say with $B < p \le 2B$ for some given B, we simply test uniformly selected random numbers p in the range for primality and return the first number that passes k such tests, for some given k. On any composite number, the tests return "probably prime" with probability at most 2^{-k}. One might then want to conclude that the output is prime with probability at least $1 - 2^{-k}$. This is fallacious. Imagine that there were only few primes between B and $2B$, say just one. Then for small k one would be much more likely to receive a composite number than a prime. Thus the density of the primes enters the following result.

THEOREM 18.8.
Given positive integers B, k*, the output of the above procedure is prime with probability at least* $1 - 2^{-k+1} \ln B$*. It uses an expected number of* $O(k(\log^2 B)\mathsf{M}(\log B))$ *word operations.*

PROOF. The probability space here is the set of all random choices within the procedure. By the prime number theorem, the set P of primes considered has size

$$\#P = \pi(2B) - \pi(B) \ge \frac{B}{\ln B}\left(1 - \frac{3}{\ln B}\right) \ge \frac{B}{2\ln B}, \tag{1}$$

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provided that $B \geq 6$ (Exercise 18.18). Thus a random integer between $B+1$ and $2B$ is prime with probability at least

$$\frac{\#P}{B} \geq \frac{1}{2\ln B}.$$

We denote by C and T the events that the chosen random number is composite and that all k tests answer "probably prime", respectively. Then $\text{prob}(p \text{ prime}) \leq \text{prob}(T)$, and, with the conditional probability $q = \text{prob}(C \cap T)/\text{prob}(T)$ (Section 25.6), we have

$$\begin{aligned}(2\ln B)^{-1} \cdot q \;&\leq\; \text{prob}(p \text{ prime}) \cdot q \;\leq\; \text{prob}(T) \cdot q \\ &= \text{prob}(C \cap T) \;\leq\; \text{prob}(C \cap T)/\text{prob}(C) \;\leq\; 2^{-k},\end{aligned}$$

which implies the probability estimate. We expect to make at most $2\ln B$ choices. By Theorem 18.6, each choice costs $O(k \log B \cdot \mathsf{M}(\log B))$ word operations, and the claim follows for $B \geq 6$. If $B \in \{1, \ldots, 5\}$, then we modify the procedure to choose one of the primes $2, 3, 5$, or 7. □

The procedure can be viewed as a computationally efficient version of **Bertrand's postulate** which says that there is always a prime between an integer and its double. The estimate (1) is slightly too pessimistic for larger values of B; a more realistic assumption is $\#P \approx B/\ln B$.

In Section 6.9, we proved that a nonzero polynomial probably takes a nonzero value at a random point. We now give an integer version of this important tool for modular algorithms. Evaluating a polynomial at a point and taking an integer modulo a prime are analogous operations. If we rewrite "$f(u) = 0$" as "$f \equiv 0 \bmod x - u$" in the case $n = 1$, the integer analog of Lemma 6.44 reads as follows.

LEMMA 18.9. *Let $P \subseteq \mathbb{N}$ be a finite nonempty set of prime numbers, $a = \min P$, and $M \in \mathbb{Z}$ with $0 \neq |M| \leq C$. If p is chosen from P uniformly at random, then*

$$\text{prob}\{M \equiv 0 \bmod p : p \in P\} \leq \frac{\log_a C}{\#P}. \tag{2}$$

PROOF. There are at most $\log_a |M| \leq \log_a C$ primes in P that divide M. □

We are now in a position to provide one of the ingredients for the modular algorithms for the determinant (Section 5.5), gcds and the Extended Euclidean Algorithm (Chapter 6), root finding (Sections 14.5 and 15.6), and factorization (Chapters 15 and 16) in $\mathbb{Z}[x]$, and for cryptography: finding suitable primes. Table 18.1 summarizes the costs and requirements. The second last algorithm does not occur explicitly in Section 16.5; essentially, one has to replace the first four steps of

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	modular algorithm	prime requirements	cost for prime finding	cost for algorithm
determinant	big prime §5.5	$p > 2n^{n/2}A^n$	$n^3 \log^3 A$	$n^4 \log A$
	small primes 5.10	$p_1, \ldots, p_r < 2r \ln r$	$n \log A$	$n^4 \log A$
gcd	big prime 6.34	$p > \sqrt{n+1} \cdot 2^{n+1} A^2$	$n^3 + \log^3 A$	$n^2 + n \log A$
	small primes 6.38	$p_1, \ldots, p_r < 2r \ln r$	$n \log A$	$n^2 + n \log A$
EEA	small primes 6.57	$p_1, \ldots, p_r < 2r \ln r$	$n \log A$	$n^3 \log A$
root finding	big prime 14.17	$p > 2n(A^2 + A)$	$\log^3 A$	$n^2 \log A + n \log^2 A$
	prime power §15.6	$p < 2r \ln r$	$n \log A$	$n^2 \log A$
factorization	big prime 16.22	$p > \sqrt{n+1} \cdot 2^{n+1} A^2$	$n^3 + \log^3 A$	$n^{10} + n^8 \log^2 A$
	prime power 16.22	$p < 2r \ln r$	$n \log A$	$n^{10} + n^8 \log^2 A$

TABLE 18.1: Costs and requirements of various modular algorithms on inputs of degree (or dimension) n and max-norm at most A. For all small prime and prime power algorithms, there is a parameter $r \in O(n \log(nA))$. For some big prime algorithms, we also have the requirement that p does not divide a certain subresultant, of word length $O(n \log(nA))$. The last column contains the running time for the remaining algorithm without the prime finding stage. All stated costs are with fast arithmetic and ignore logarithmic factors.

Algorithm 16.22 by the first three steps of the big prime algorithm 15.2. For most algorithms, we see in the fourth column that the time for finding one or several small primes is much less than the time for finding a big prime; in practice, we would work with a precomputed list of small primes, as discussed below. However, also the remaining stages of small primes and prime power algorithms are faster than the corresponding stages of their big prime counterparts: in theory only by logarithmic factors, which do not show up in the last column of Table 18.1, but they are clearly visible in practice (see Section 6.13).

THEOREM 18.10.

(i) There is a probabilistic algorithm which, with probability at least $3/4$, returns a prime p between $B+1$ and $2B$, for any positive integer $B \in \mathbb{N}$ of word length β. Moreover, if $M \in \mathbb{Z}$ is a nonzero number such that $6 \ln |M| \leq B$, then p is prime and p does not divide M with probability at least $1/2$. The algorithm takes $O(\beta^2 \cdot \mathsf{M}(\beta) \log \beta)$ word operations.

(ii) For $r \in \mathbb{N}$, we can compute the first r prime numbers $p_1 = 2, \ldots, p_r \in \mathbb{N}$ deterministically at a cost of $O(r \log^2 r \log\log r)$ word operations, and each of them is less than $2r \ln r$ if $r \geq 2$.

PROOF. (i) If $B \geq 6$, then Theorem 18.8 with $k = 2 + \lceil \log_2 \ln B \rceil$ gives the first claim. Using Lemma 18.9 and (1), we find that p divides M with probability

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at most

$$\frac{\log_B C}{\#P} \leq \frac{\ln C \cdot 2\ln B}{\ln B \cdot B} \leq \frac{1}{3}$$

if it is prime. Therefore the probability that p has the required properties is at least $\frac{3}{4}(1-\frac{1}{3}) = \frac{1}{2}$. If $B \in \{1,\ldots,5\}$, then $|M| \leq \lfloor e^{B/6} \rfloor \leq B$, so that none of the primes between $B+1$ and $2B$ divides M, and we may take one of $2,3,5$, or 7 for p.

(ii) We have $p_r < r(\ln r + \ln\ln r - 1/2) \leq 2r\ln r$ for $r \geq 20$, by the prime number theorem 18.7. In fact, $p_r < 2r\ln r$ for all $r \geq 2$. We find our primes by the **sieve of Eratosthenes**, as follows. We write down a list of all integers below $x = 2r\ln r$. Then we cross out all even numbers, all multiples of 3, all multiples of 5, and so on, for each prime less than $\sqrt{x}$. The remaining integers are not divisible by a prime less than $\sqrt{x}$, and hence they are prime. The cost is $\lfloor x/p \rfloor$ steps for each prime $p \leq \sqrt{x}$, altogether at most

$$x \sum_{\substack{p < \sqrt{x} \\ p \text{ prime}}} \frac{1}{p} \in O(x \operatorname{loglog} x)$$

steps, by equation (3.20) in Rosser & Schoenfeld (1962). We may implement a single step by setting a flag for "crossed out" and incrementing a counter by p, taking $O(\log x)$ word operations. We obtain a total cost of $O(x\log x\operatorname{loglog} x)$ word operations, and the claim follows from $\log x \in O(\log r)$. □

Our first application is the big prime modular gcd algorithm 6.34.

COROLLARY 18.11.
Let $n \in \mathbb{N}_{\geq 2}$, $f,g \in \mathbb{Z}[x]$ be primitive, with degrees at most n and max-norms at most A, $h = \gcd(f,g)$, $b = \gcd(\operatorname{lc}(f),\operatorname{lc}(g))$, $B = \lceil (n+1)^{1/2}2^{n+1}bA \rceil$, and $\beta = \log B$. If $n \geq 5$ or $A \geq 5$, then we can find an integer p with $B < p \leq 2B$ such that, with probability at least $1/2$, p is prime and does not divide $\operatorname{res}(f/h, g/h)$. This algorithm uses $O(\beta^2 \mathsf{M}(\beta)\log\beta)$ or $O^{\sim}(n^3 + \log^3 A)$ word operations.

PROOF. We let $\sigma = \operatorname{res}(f/h, g/h)$, and have $|\sigma| \leq (n+1)^n A^{2n}$, by Theorem 6.35, and therefore $6\ln|\sigma| \leq 12n\ln((n+1)A)$. Since $12n < 2^{n+1}$ and $\ln((n+1)A) < (n+1)^{1/2}A$ if $n \geq 5$, and

$$12n\ln((n+1)A) < (n+1)^{1/2}2^{n+1}A \leq B$$

if $n = 2,3,4$, for all $A \geq 5$, we find $6\ln|\sigma| \leq B$ in all cases, and the claims follow from Theorem 18.10 (i) and $\beta \in O(n + \log A)$. □

It is satisfying to have a worst-case guarantee on the cost of prime generation, and to see yet another reason why the big prime modular algorithms are impractical

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in spite of their conceptual simplicity. In Figure 6.4 the big prime running times are quite erratic, but smooth when the cost of prime generation is suppressed. In small primes or prime power modular algorithms, the cost of obtaining primes is negligible.

The prime finding step in the factorization algorithms 15.2 (big prime) and 15.19 and 16.22 (prime power) in $\mathbb{Z}[x]$ is quite inexpensive.

COROLLARY 18.12.
Let $f \in \mathbb{Z}[x]$ be squarefree of degree $n \geq 2$ and with max-norm $\|f\|_\infty = A$, $\gamma = 2n\ln((n+1)A)$, and suppose that $A \geq 5$ if $n \leq 4$.

(i) There is a probabilistic algorithm which, with probability at least $1/2$, outputs a prime p between $B+1$ and $2B$ and not dividing $\operatorname{res}(f, f')$, where $B = \lceil (n+1)^{1/2} 2^{n+1} |\operatorname{lc}(f)| A \rceil$. Then $\beta = \log B \in O(n + \log A)$, and the algorithm uses an expected number of $O(\beta^2 \mathsf{M}(\beta) \log \beta)$ or $O^\sim(n^3 + \log^3 A)$ word operations.

(ii) With $O(\gamma \log^2 \gamma \operatorname{loglog} \gamma)$ or $O^\sim(n \log A)$ word operations we can find (probabilistically) a prime p of word length in $O(\log \gamma)$ and such that $p \nmid \operatorname{res}(f, f')$ with probability at least $1/2$.

PROOF. (i) We let $\sigma = \operatorname{res}(f, f')$ and $C = (n+1)^{2n} A^{2n-1}$. As in the proof of Corollary 18.11, we have $6 \ln C \leq 6\gamma \leq B$. Since $\|f'\|_\infty \leq nA$ and $0 < |\sigma| \leq C$ by Theorem 6.23, the claim follows from Theorem 18.10 (i).

(ii) We let σ and C be as in the proof of (i). Then Theorem 18.10 (ii) with $r = \lceil 2 \log_2 C \rceil \in O(\gamma)$ says that we can compute the first r primes, each less than $2r \ln r$, within the stated time bound. By the choice of r, at most half of these primes divide σ, and we choose one of them uniformly at random. □

With some more calculations using the prime number theorem one can shave off a logarithmic factor in the estimate of (ii) (Exercise 18.21). Similar improvements are possible for small primes modular determinant computation (Algorithm 5.10), small primes modular gcd computation (Algorithm 6.38), and the small primes modular EEA (Algorithm 6.57).

A software implementation of small primes or prime power modular algorithms should precompute a table of small primes, so that for most purposes only table look-up is needed. Rather than using the first primes, it is more efficient to take the largest single precision primes. As discussed at the beginning of Chapter 5 and in Section 8.3, it is advantageous in the small primes modular approach to choose $p_1, \ldots, p_r$ to be Fourier primes, so that $p_i - 1$ is divisible by some large power 2^t of 2, for all i. There are quantitative versions of Dirichlet's (1837) famous theorem on primes in arithmetic progressions that give asymptotic estimates, but even

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the best versions (Alford, Granville & Pomerance 1994, Bach & Sorenson 1996) are considerably less precise than the prime number theorem 18.7. For practical purposes, however, it is reasonable to assume that a random number $p \equiv 1 \bmod 2^t$ near x is prime with probability about $2/\ln x$. To find enough such primes, we consecutively test 2^t+1, $2\cdot 2^t+1$, $3\cdot 2^t+1$, ... for primality until we have found r primes. This will be a precomputation stage. Exercise 18.19 estimates the number of single precision Fourier primes for 32-bit and 64-bit processors.

In our big prime algorithms for gcd computation and factorizations, it may happen that the number p given by Theorem 18.10 is not prime. If we stumble upon a nonzero element which is not invertible in our computation, then we recognize our p as composite and start all over again with a new one. However, it may happen that all computations go through even if p is composite. In the gcd case, it is possible to show that the output is nevertheless correct, but in the polynomial factoring algorithm, reducible polynomials may wrongly be declared irreducible. However, this only happens with probability at most $1/2$, by Theorem 18.10, and we may simply rerun the whole algorithm several times independently to make the error probability arbitrarily small. Or, preferably, we use the prime power factoring algorithm, which is faster and returns the complete factorization in any case.

18.5. The Solovay and Strassen test

The following two sections are not used in what follows and may be skipped at first reading. We do not give proofs.

The first probabilistic polynomial-time test for primality is due to Solovay & Strassen (1974, published 1977). The **Legendre symbol** is defined for $a, N \in \mathbb{Z}$ with N prime as

$$\left(\frac{a}{N}\right) = \begin{cases} 1 & \text{if } \gcd(a,N) = 1 \text{ and } a \text{ is a square modulo } N, \\ -1 & \text{if } \gcd(a,N) = 1 \text{ and } a \text{ is not a square modulo } N, \\ 0 & \text{if } \gcd(a,N) \neq 1, \end{cases}$$

A famous theorem of number theory, Gauß' **law of quadratic reciprocity**, says that for distinct odd primes a and N the two symbols

$$\left(\frac{a}{N}\right) \text{ and } \left(\frac{N}{a}\right)$$

are equal unless and only unless both a and N are congruent to 3 modulo 4. The **Jacobi symbol** is the generalization to an arbitrary odd N. If $N = p_1^{e_1} \cdots p_r^{e_r}$ is its prime factorization, then it is defined as

$$\left(\frac{a}{N}\right) = \left(\frac{a}{p_1}\right)^{e_1} \cdots \left(\frac{a}{p_r}\right)^{e_r}.$$

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This quantity can be computed by an efficient method, akin to the Euclidean Algorithm, without actually factoring N (Notes 18.5 and Exercise 18.23).

When N is a prime, then Lemma 14.7 implies that

$$\left(\frac{a}{N}\right) \equiv a^{(N-1)/2} \bmod N \tag{3}$$

for all $a \in \mathbb{Z}$. Solovay & Strassen (1977) prove that (3) is false for at least half of all a in $\{1,\ldots,N-1\}$ if N is composite and not a prime power. Their algorithm checks (3) for randomly chosen a; each test takes $O(\log N \cdot \mathsf{M}(\log N))$ word operations.

Although Berlekamp's (1970) probabilistic algorithm for factoring polynomials (Section 14.8) had been around for a while, it was the Solovay & Strassen (1977) result for integers that aroused widespread interest in the power of randomized algorithms. (Are numbers more intuitive to computer scientists than polynomials? The reader should by now—and even more so after reading Chapter 19—be convinced that polynomials are much easier objects than numbers.) See Notes 6.5.

18.6. Primality tests for special numbers

For some special integers of interest in number theory, particularly efficient primality tests have been designed. Pepin (1877) obtained the following primality test for the Fermat numbers $F_n = 2^{2^n}+1$ (see page 513 and Section 18.2):

$$F_n \text{ prime} \iff 3^{(F_n-1)/2} \equiv -1 \bmod F_n.$$

In fact, this is just the Solovay–Strassen test with $a = 3$ (Exercise 18.25). If p is a prime dividing F_n, then p is of the form $p = a \cdot 2^{n+2}+1$ for some integer a (Exercise 18.26). Trying such divisors led to the large factor $5 \cdot 2^{1947}+1$ of F_{1945}.

The **Mersenne numbers** are defined by $M_n = 2^n - 1$ for $n \geq 1$. The integer M_a is a nontrivial factor of M_{ab}, for $a,b \in \mathbb{N}$, and hence if M_n is prime, then so is n. There is a special test for these numbers, the *Lucas-Lehmer test* (Lucas 1878, Lehmer 1930, 1935). The test says that 2^n-1 is prime if and only if $l_{n-1} \equiv 0 \bmod 2^n-1$, where l_i is recursively defined by $l_1 = 4$ and $l_i = l_{i-1}^2 - 2$ for $i \geq 1$.

Presently it is not known if there are infinitely many Fermat primes or infinitely many Mersenne primes. No new Fermat primes have been discovered since Fermat's time, while there are now 47 Mersenne primes known, as mentioned at the beginning of this chapter.

Williams & Dubner (1986) used a Lucas-Lehmer type test to prove the primality of the **repunit**

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$$\frac{10^{1031}-1}{9} =$$

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the criterion is true. In his 11-line reply, Korselt (1899) gives the definition of Carmichael numbers (as on page 520) and states that he has proved the characterization of Exercise 18.9 (ii), namely squarefreeness and divisibility of $N-1$ by the least common multiple of all $p-1$. Thus we may consider Korselt as the discoverer of Carmichael numbers, and the characterization is known as **Korselt's criterion**. The editor notes after Korselt's reply that five other replies were received to this question; *toutes à peu près dans le même sens*[1].

Lenstra (1979b) proves that if $a^{N-1} \equiv 1 \bmod N$ for every prime $a < \ln^2 N$, then N is squarefree.

18.3. Miller (1976) proposed a deterministic version of the strong pseudoprimality test 18.5 and showed that it runs in polynomial time under the ERH (Theorem 18.6), and Rabin (1976, 1980a) suggested the probabilistic variant. Neither of the two algorithms looks for factors of Carmichael numbers. Earlier versions of the test were given by Dubois (1971) and Selfridge (not later than 1974, unpublished), but they did not reach a wide audience. Dubois suggests the strong pseudoprimality test with $a = 2, 3$, and 5. He is well aware that it may fail, and proposes to use $a = 7$ as well. See the end of these Notes for the smallest N on which this variant fails.

Bach, Miller & Shallit (1986) state a generalized version of Algorithm 18.5 for integer factorization, without actually mentioning the application to the Carmichael case. The fact that Carmichael numbers can be factored in random polynomial time seems to be folklore. Alford, Granville & Pomerance (1994) solved a long-standing open problem by proving that there are infinitely many Carmichael numbers.

For an odd composite integer N, the probability for a random $a \in \{1, \ldots, N-1\}$ with $\gcd(a, N) = 1$ to be a strong liar is at most $1/2$, by the proof of Theorem 18.6. Rabin (1980a), Monier (1980), and Atkin & Larson (1982) have shown the smaller bound $1/4$. To generate random primes, suppose that we fix n and k, choose n-bit odd numbers uniformly at random, subject them to k strong pseudoprimality tests, and return the first one that passes all these tests. We call $p_{n,k}$ the probability that a composite number is returned. Damgård, Landrock & Pomerance (1993) deal with the subtleties of estimating $p_{n,k}$, as noted before Theorem 18.8, and prove several estimates, for example:

$$p_{600,1} \le 2^{-75}, \quad p_{n,k} < \frac{1}{7}\, n^{15/4}\, 2^{-n/2-2k} \text{ if } 4k \ge n \ge 21.$$

All reasonably small numbers have small strong witnesses: Pomerance, Selfridge & Wagstaff (1980) prove that for all composite $N \le 25 \times 10^9$ (except for $N = 3\,215\,031\,751$), at least one of 2, 3, 5, and 7 is a strong witness. Pinch (1993) describes erroneous results of primality tests implemented in some computer algebra systems.

Both the Solovay–Strassen and the strong pseudoprimality tests should properly be called **compositeness tests** because they show the set PRIMES of all prime numbers is in co-$\mathcal{RP}$ and its complement (without 0 and 1) COMPOSITES is in $\mathcal{RP}$, which is a (possibly proper) subset of $\mathcal{BPP}$ (Section 25.8), but the wrong terminology has stuck.

18.4. The prime number theorem is a central result in number theory, and has a long and distinguished history. Proofs of the asymptotic version stated first in Theorem 18.7 are in many texts, for example Hardy & Wright (1985). The precise version stated in Theorem 18.7 is from Rosser & Schoenfeld (1962).

[1] all in about the same sense

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An early attempt at the prime number theorem was by Legendre (1798), and Gauß (1849) said that he found the estimate around 1792. Chebyshev (1849, 1852) proved that $\pi(x)$ is asymptotically $x/\ln x$, up to a constant factor, and de la Vallée Poussin (1896) and Hadamard (1896) proved that $\pi(x) = x/\ln x + o(x/\ln x)$. A better approximation is given by the **logarithmic integral** $\pi(x) \approx \mathrm{Li}(x) = \int_2^x dt/\ln t$.

A vital tool in modern prime number theory is **Riemann's** (1859) **zeta function** $\zeta(s)$, a meromorphic function on the complex plane. It is obtained by analytic continuation of

$$\zeta(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1} = \sum_{n \geq 1} n^{-s},$$

which is defined when $\Re s > 1$, and which already Euler used. Riemann made his famous conjecture, the **Riemann Hypothesis**, that all zeroes s of ζ lie on the **critical line** $\Re s = 1/2$: [...] *und es ist sehr wahrscheinlich, dass alle Wurzeln* [von $\zeta(\frac{1}{2} + it)$] *reell sind. Hiervon wäre allerdings ein strenger Beweis zu wünschen; ich habe indess die Aufsuchung desselben nach einigen flüchtigen vergeblichen Versuchen vorläufig bei Seite gelassen, da er für den nächsten Zweck meiner Untersuchung entbehrlich schien.*[2] A proof of this conjecture, still elusive after over 130 years, would imply dramatic improvements in the estimates for the error term in the prime number theorem. Clever methods have been devised to calculate billions of roots of the zeta function (van de Lune, te Riele & Winter 1986, Odlyzko & Schönhage 1988, Odlyzko 1995c); fast arithmetic is a must for such high-performance calculations.

Already Legendre (1798) had used a logarithmic integral, but he was well aware that he had no proof of his (incorrect) formula. He set the task solved in this chapter: *Il serait à désirer, pour la perfection de la théorie des nombres, qu'on trouvât une méthode praticable au moyen de laquelle on pût décider assez promptement si un nombre donné est premier ou s'il ne l'est pas.*[3] Did he already feel that there is a computational difference between testing primality and factoring?

The zeta function has been generalized from integers to algebraic number fields. The conjecture that all those generalizations have their zeroes on the critical line is called the Extended Riemann Hypothesis. For several algorithms, the estimates of their running time (or their proofs of correctness) rely on the ERH; see Notes 14.9.

Pritchard (1983, 1987) and Sorenson (1998) give several more efficient versions of the sieve of Eratosthenes.

18.5. We heard the "unless and only unless" from Hendrik Lenstra. The "iff" was coined by Halmos (see Halmos 1985, page 403), and Conway invented comic imitations like "unlesss".

Monier (1980) compares the two tests by Solovay & Strassen (1977) and by Miller and Rabin. Eisenstein (1844) and Lebesgue (1847) present algorithms for the Jacobi symbol. They are analyzed in Shallit (1990), and efficient methods are given by Bach & Shallit (1996), §5.9, and Meyer & Sorenson (1998).

Miller, Rabin, Solovay, and Strassen shared the ACM Paris Kanellakis Award for their work.

[2] [...] and it is very probable that all roots [of $\zeta(\frac{1}{2} + it)$] are real. A rigorous proof of this would be desirable; I have, however, left aside the quest for one after several brief and unsuccessful attempts, since it seemed dispensable for the immediate goal of my investigation.

[3] It would be desirable for the perfection of number theory to find a practical method by which one should be able to decide fairly quickly whether a given number is prime or not.

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18.6. See Bach & Shallit (1996) for details on Pepin's test, and Hardy & Wright (1985), §2.5, for the example F_5. In fact, Pepin used 5 instead of 3 as his witness. It is widely conjectured that no $F_n > F_4$ is prime, but Wagstaff (1983) has conjectured that

$$\#\{p < x \mid M_p \text{ is prime}\} \approx \frac{e^\gamma}{\ln 2} \operatorname{lnln} x \approx 2.57 \operatorname{lnln} x,$$

where $\gamma = 0.5772156649\ldots$ is Euler's constant. In particular, this would imply that there are infinitely many Mersenne primes. The term "Mersenne number" was apparently coined by Rouse Ball in 1892 (see Ball & Coxeter 1947, page 65: *Mersenne's Numbers*), and "repunit" is from Beiler (1964), page 83.

Further notes. The largest **twin primes** known at the time of writing are $1\,807\,318\,575 \cdot 2^{98305} \pm 1$, with 29 603 decimal digits, discovered by Underbakke, Carmody, and Gallot. Caldwell's prime list `http://www.utm.edu/research/primes` regularly updates this and many other prime records. These numbers have a very special form and can be proven to be prime by deterministic methods which are far too slow for general numbers of this size.

Exercises.

18.1 Show that $a^2 + b^2 \equiv 0, 1$, or 2 modulo 4 for all $a, b \in \mathbb{Z}$.

18.2 Compute $2^{1\,000\,005} \bmod 55$. Hint: This needs virtually no calculation.

18.3 Which of the two integers $10^{200} + 349$ and $10^{200} + 357$ is probably prime and which is certainly composite? You may use a computer algebra system to find this out, but you should not use routines like `isprime` or `ifactor`. Warning: not every exponentiation routine is suited for solving this task.

18.4* You are to determine precisely the error probability of the Fermat test in a special case. Let $p \neq q$ be primes with $p \equiv q \equiv 3 \bmod 4$ and $\gcd(p-1, q-1) = 2$, and $N = pq$.

(i) Prove that $\gcd(N-1, p-1) = 2$, and conclude that $\{u^{N-1} : u \in \mathbb{Z}_p^\times\} = \{u^2 : u \in \mathbb{Z}_p^\times\}$ and $\operatorname{prob}(a^{N-1} \equiv 1 \bmod p) = 2/(p-1)$ for a uniform random element $a \in \{1, \ldots, p-1\}$. Hint: Exercise 14.11.

(ii) Calculate the probability that the Fermat test outputs "possibly prime" on input N, assuming that a is chosen from $\{1 \leq c < N : \gcd(c, N) = 1\}$ uniformly at random in step 1. Compare your result numerically to the estimate from Theorem 18.3 for $p = 79$ and $q = 83$.

18.5* (i) Find all Fermat liars for $N = 15$.

(ii) Show that if p and $2p-1$ are both prime and $N = p(2p-1)$, then 50% of the elements in $\mathbb{Z}_N^\times$ are Fermat liars, namely all those which are squares modulo $2p-1$.

18.6 (Lenstra, Lenstra, Manasse & Pollard 1993) Let N be composite.

(i) Let $a \in \{1, \ldots, N-1\}$ with $\gcd(a, N) = 1$ be a Fermat witness and $g = \gcd(a^{N-1} - 1, N)$. Prove that N is not a prime power if $g = 1$.

(ii) State and prove a similar criterion when a is a strong witness.

18.7 Prove that for $N = 3\,215\,031\,751$, the smallest strong witness is 11.

18.8 Find a 20 decimal digit prime. Explain how you obtained it and why you believe it is prime. You may find functions such as MAPLE's `isprime` useful.

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18.9 Let $N \in \mathbb{N}_{>3}$. Prove:

(i) If N is a Carmichael number, then it has the property that $p-1$ divides $N-1$ for all prime divisors p of N. Hint: CRT and Exercise 8.16.

(ii) N is prime or a Carmichael number if and only if it is squarefree and the property in (i) holds.

(iii) A Carmichael number is odd and has at least three distinct prime divisors.

(iv) Which of the following integers are Carmichael numbers: 561, 663, 867, 935, 1105, 1482, 1547, 1729, 2077, 2465, 2647, 2821, 172081? You may use the integer factoring routine of your favorite computer algebra system.

18.10* (i) Find all Carmichael numbers of the form $3pq$, where $p \neq q$ are primes.

(ii) Find all Carmichael numbers of the form $5pq$, where $p \neq q$ are primes.

(iii) Show that for any fixed prime number r there are only finitely many Carmichael numbers of the form rpq, with distinct primes p, q.

18.11* For $N \in \mathbb{N}_{\geq 2}$, we denote by $D(N) = \{1 \leq d < N : d \mid N\}$ the set of proper divisors of N.

(i) Let $n \in \mathbb{N}$ be such that $M_n = 2^n - 1$ is a Mersenne prime. Show that $P_n = 2^{n-1}M_n$ is a **perfect number**, so that

$$\sum_{d \in D(P_n)} d = P_n.$$

(These are the only even perfect numbers, and it is unknown whether there are odd perfect numbers.)

(ii) Assume that $p_1 = 6m+1$, $p_2 = 12m+1$, and $p_3 = 18m+1$ are primes, for some $m \in \mathbb{N}_{\geq 1}$. Prove that $p_1p_2p_3$ is a Carmichael number. (The smallest example is $1729 = (6+1)(12+1)(18+1)$, where $m = 1$. Can you find other examples? $1729 = 1 + 12^3 = 9^3 + 10^3$ is Hardy and Ramanujan's famous **taxi-cab number**, the smallest positive integer which is the sum of two cubes in two different ways; see Hardy 1937, page 147.)

(iii) Let $D(P_n) = \{d_1, d_2, \ldots, d_{2n-1}\}$, $m \in \mathbb{N}_{\geq 1}$, $p_i = d_i P_n m + 1$ for $1 \leq i < 2n$, and $N = p_1 \cdots p_{2n-1}$. Show that $p_i - 1$ divides $N-1$ for $1 \leq i < 2n$, and conclude that N is a Carmichael number if $p_1, \ldots, p_{2n-1}$ are prime. (Part (ii) is the special case $n = 2$.)

18.12* (i) Modify Algorithm 18.5 so as to find all prime factors of a Carmichael number N. Your algorithm should take a confidence parameter $k \in \mathbb{N}$ as additional input, such that each factor in the output is prime with probability at least $1 - 2^{-k}$, and it should use an expected number of $O((k + \log N)\log N \cdot \mathsf{M}(\log N))$ word operations.

(ii) Generalize your algorithm from (i) so that it can factor a squarefree number N if a multiple L of $\lambda(N)$ (see Exercise 18.13) is known. Prove the same time bound as for (i) if $\log L \in O(\log N)$. (A more general algorithm is in Bach, Miller & Shallit 1986).

18.13 Let $N > 1$ be an odd integer with prime factorization $N = p_1^{e_1} \cdots p_r^{e_r}$. The **Carmichael function** $\lambda(N)$ is defined as the least common multiple of $\varphi(p_1^{e_1}), \ldots, \varphi(p_r^{e_r})$, where φ is Euler's totient function.

(i) Prove that $a^{\lambda(N)} = 1$ for all $a \in \mathbb{Z}_N^\times$.

(ii) Prove that $a^{N-1} = 1$ for all $a \in \mathbb{Z}_N^\times$ if and only if $\lambda(N) \mid N-1$. Hint: Exercise 18.9.

18.14* Let $N > 1$ be an odd integer, $\lambda(N)$ as in Exercise 18.13, and $C_N = \{a \in \mathbb{Z}_N^\times : a^{\lambda(N)/2} = \pm 1\}$.

(i) Prove that C_N is a multiplicative subgroup of $\mathbb{Z}_N^\times$.

(ii) Show that if $N = p^e$ for some $e \geq 1$ and some prime p, then $C_N = \mathbb{Z}_N^\times$. Hint: Exercise 9.40.

(iii) Prove the converse of (ii). Hint: CRT.

(iv) Recall that N is a perfect power if $N = m^l$ for some integers $m, l > 1$. Discuss whether the following is a good primality testing algorithm: First check whether N is a perfect power. If it is not, then output "probably prime" if $a^{\lambda(N)/2} = \pm 1$ for a randomly chosen $a \in \mathbb{Z}_N^\times$, and return "composite" otherwise.

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18.15 Let $p \in \mathbb{N}$ be an odd prime.

(i) Prove that no square in $\mathbb{F}_p^\times$ is a generator of that group.

(ii) If $p = 2^{2^n} + 1$ is a Fermat prime, for some $n \in \mathbb{N}$, then show that any nonsquare in $\mathbb{F}_p^\times$ is a generator of that group. Hint: Exercise 8.16.

18.16 Let $r, s \in \mathbb{N}_{\geq 1}$ such that $p = r2^s + 1$ is prime.

(i) Let $a \in \mathbb{N}$ be a nonsquare modulo p. Prove that $a^r \bmod p$ is a primitive 2^sth root of unity in $\mathbb{F}_p^\times$. Hint: Exercise 8.15.

(ii) Design a probabilistic algorithm that on input r, s computes a primitive 2^sth root of unity in $\mathbb{F}_p^\times$ (the algorithm need not check whether p is prime), and prove that it uses an expected number of $O(\log p \cdot \mathsf{M}(\log p))$ word operations.

(iii) Using a computer algebra system, find all primes p between 2^{31} and 2^{32} such that $p - 1$ is divisible by 2^{27}, and find a primitive 2^{27}th root of unity modulo each such prime. Prove that 27 is the largest exponent for which three such primes exist.

18.17 Show that for each positive integer n, there exists a positive integer a such that $a, a+1, a+2, \ldots, a+n$ are all composite.

18.18 (i) Prove that $\pi(2x) - \pi(x) > x/(2\ln x)$ if $x \geq e^6$. Hint: Prime number theorem.

(ii) Write a program which checks that the claim from (i) is indeed true for $x \geq 6$.

(iii) How many primes between 2^{k-1} and 2^k are there at least for $k = 32$ and $k = 64$?

(iv) Assuming that $n \approx \log_2 B$ in the small primes modular determinant algorithm 5.10, for which values of n are k-bit primes sufficient for $k = 32$ and $k = 64$?

18.19 Let k and s be integers with $0 < s < k$. A **single precision Fourier prime** is a prime p with $2^{k-1} < p < 2^k$ and $p \equiv 1 \bmod 2^s$. For simplicity, you may assume that the probability for a random number of the form $1 + q2^s$ between 2^{k-1} and 2^k to be prime is $2/(k-1)\ln 2$, as discussed at the end of Section 18.4.

(i) For $k = 32$ and $k = 64$ and $1 \leq s < k$, estimate the number of single precision Fourier primes and tabulate the estimates. Compare them to the actual number of such primes for $k - 8 \leq s < k$.

(ii) Consider the small primes modular multiplication algorithm in $\mathbb{Z}[x]$ from Section 8.4. Use your estimates from (i) to answer the following question for processors with word size $k = 32$ and $k = 64$. Assuming that you want to multiply polynomials of degree less than 2^{s-1} with 2^{s-1}-bit coefficients by that method using single precision Fourier primes, what is the maximal possible value of s?

18.20* The following method improves the cost in Theorem 18.8 by a logarithmic factor, by first testing for small prime factors. We use the fact that the number of integers up to x having no prime factor less or equal to y is asymptotically at most $x/\ln y$ (see Tenenbaum (1995), Theorem III.6.3).

Given B and k as for Theorem 18.8, we set $\beta = \lceil \ln B \rceil$ and let $r \in \mathbb{N}$ be a parameter. We choose integers $p \in \{B+1, \ldots, 2B\}$ uniformly at random, use Pollard & Strassen's algorithm 19.2 to check whether p has a prime factor less or equal to r, and if not, then we run k strong pseudoprimality tests on p. We return the first p that is always pronounced "probably prime". Prove:

(i) The expected cost for all smoothness tests is $O(\beta \mathsf{M}(r^{1/2}) \mathsf{M}(\beta)(\log r + \log \beta))$.

(ii) The output is prime with probability (over the random choices) at least $1 - 2^{-k+1}\beta/\ln r$.

(iii) If $\mathsf{M}(\beta) \in O(\beta^2/\log^2 \beta)$ and we choose $r = \beta$, then the analog of Theorem 18.8 is valid with success probability at least $1 - 2^{-k+1}\beta/\ln \beta$ and cost $O(k\beta^2 \mathsf{M}(\beta)/\log \beta)$.

18.21* You are to improve the time estimates for some small primes and prime power modular algorithms over $\mathbb{Z}$ by logarithmic factors. The function $\vartheta(x) = \sum_{p \leq x} \ln p$, with p running over the primes, will be useful. We have $\vartheta(x) \approx x$, and Rosser & Schoenfeld (1962) show more precisely that

$$x\left(1 - \frac{1}{\ln x}\right) < \vartheta(x) < x\left(1 + \frac{1}{2\ln x}\right) \text{ if } x \geq 41.$$

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(i) The small primes modular determinant algorithm 5.10 and the small primes modular EEA 6.57 only require that we find a collection of primes $p_1, \ldots, p_r \leq x$ such that their product exceeds a given bound $C \in \mathbb{R}_{>0}$, and we took the first $r \approx \log_2 C$ primes for simplicity (Theorems 5.12 and 6.58), so that $x = p_r \approx (\log_2 C) \ln \log_2 C$. But in fact, $\vartheta(x) \geq \ln C$ is sufficient, which leads to the choice $x \approx \ln C$, by the above, so that $r = \pi(x) \approx \ln C / \ln \ln C$. Work out the details and show that the cost of Algorithms 5.10 and 6.57 drops to $O(n^4 \log(nB) \operatorname{loglog}(nB) + n^3 \log^2(nB))$ and $O(n^3 m \log^2(nA))$ word operations, respectively.

(ii) In Corollary 18.12 (ii), the requirement is slightly different: we need r primes such that the product of each $r/2$ of them exceeds the discriminant bound C. Thus we may take $x \approx \ln C$ and $r = 2\pi(x) \approx 2 \ln C / \ln \ln C$. Use this to improve the cost estimate of Corollary 18.12 (ii) to $O(\gamma \log \gamma \operatorname{loglog} \gamma)$ word operations.

18.22 Let $p \in \mathbb{N}$ be an odd prime.

(i) Prove that 4 divides $p-1$ if -1 is a square modulo p. Hint: Lagrange's theorem.

(ii) Prove the converse of (i). Hint: Consider $a^{(p-1)/4}$ for a nonsquare $a \in \mathbb{F}_p^\times$.

(iii) Conclude that the Legendre symbol $(\frac{-1}{p})$ is 1 if and only if $p \equiv 1 \bmod 4$.

18.23* (i) Show that the Jacobi symbol is multiplicative with respect to both arguments:

$$\left(\frac{ab}{N}\right) = \left(\frac{a}{N}\right)\left(\frac{b}{N}\right), \quad \left(\frac{a}{MN}\right) = \left(\frac{a}{M}\right)\left(\frac{a}{N}\right)$$

for all $a, b, M, N \in \mathbb{N}_{>0}$ with $M, N \geq 3$ odd.

(ii) Prove that the law of quadratic reciprocity also holds for the Jacobi symbol: If $a, N \in \mathbb{N}$ are coprime and odd, then $(\frac{a}{N})$ and $(\frac{N}{a})$ are equal unless and only unless $a \equiv N \equiv 3 \bmod 4$.

(iii) A special case of the law of quadratic reciprocity is that $(\frac{2}{N}) = 1$ if and only if $N \equiv \pm 1 \bmod 8$ for an odd prime $N \in \mathbb{N}$. Prove that this also holds for the Jacobi symbol, where $N \geq 3$ is an arbitrary odd integer.

(iv) Show that $(\frac{a}{N}) = (\frac{a \operatorname{rem} N}{N})$ for all $a, N \in \mathbb{N}_{\geq 1}$ with $N \geq 3$ odd.

(v) Write an efficient algorithm that, given an odd integer $N > 1$ and $a \in \{1, \ldots, N-1\}$, computes the Jacobi symbol $(\frac{a}{N})$, and analyze its cost.

18.24* (Lehmann 1982) Let $N \in \mathbb{N}_{\geq 3}$ be odd, $\sigma : \mathbb{Z}_N^\times \longrightarrow \mathbb{Z}_N^\times$ the power map $\sigma(a) = a^{(N-1)/2}$, and $T = \operatorname{im}(\sigma) \subseteq \mathbb{Z}_N^\times$.

(i) Show that $T = \{1, -1\}$ if N is prime.

(ii) Prove that $T \neq \{1, -1\}$ if N is not a prime power. Hint: Assume that $-1 \in T$ and apply the Chinese Remainder Theorem.

(iii) Show that $T \neq \{1, -1\}$ if $N = p^e$ for a prime $p \in \mathbb{N}$ and $e \in \mathbb{N}_{\geq 2}$. Hint: Lemma 18.1.

(iv) Prove that N is a Carmichael number if $T = \{1\}$.

(v) Consider the following algorithm.

ALGORITHM 18.13 Lehmann's primality test.

Input: An odd integer $N \geq 3$ and a parameter $k \in \mathbb{N}$.
Output: Either "probably composite" or "probably prime".

1. **for** $i = 1, \ldots, k$ **do**
2. choose $a_i \in \{1, \ldots, N-1\}$ uniformly at random
3. **call** the repeated squaring algorithm 4.8 to compute $b_i = a_i^{(N-1)/2} \operatorname{rem} N$.
4. **if** $\{b_1, \ldots, b_k\} \neq \{1, -1\}$ **then return** "probably composite" **else return** "probably prime"

Prove that the algorithm outputs "probably prime" with probability at least $1 - 2^{1-k}$ if N is prime, and that it outputs "probably composite" with probability at least $1 - 2^{-k}$ if N is composite.

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(vi) Prove that Lehmann's algorithm can be executed with $O(k \log N \cdot \mathsf{M}(\log N))$ word operations.

(vii) Discuss the following modification of step 4: if $b_i = -1$ for $1 \leq i \leq k$, then the algorithm should return "probably prime" as well.

(viii) For each of the composite numbers $N = 343$, 561, 667, and 841, compute T and determine exactly the error probability of Lehmann's algorithm for $k = 10$ (you may assume that $\gcd(a_i, N) = 1$ for all i). Compare your results to the estimate from (v).

18.25* Let $F_n = 2^{2^n} + 1$ be the nth Fermat number, for $n \in \mathbb{N}$.

(i) Assume that F_n is prime. Show that 3 and 7 are nonsquares modulo F_n if $n \geq 1$ and that 5 is a nonsquare modulo F_n if $n \geq 2$. Hint: Exercise 18.23.

(ii) Conclude that for $n \geq 1$, Pepin's (1877) test works correctly: F_n is prime if and only if $3^{(F_n-1)/2} \equiv -1 \bmod F_n$.

(iii) Show that Pepin's test can be performed with $O(2^n \mathsf{M}(2^n))$ word operations.

18.26* Let $n \in \mathbb{N}_{\geq 2}$, $F_n = 2^{2^n} + 1$ the nth Fermat number, and $p \in \mathbb{N}$ a prime divisor of F_n. Prove that $2^{n+2} \mid p - 1$. Hint: Lagrange's theorem and Exercise 18.23.

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Problema, numeros primos a compositis dignoscendi,
hosque in factores suos primos resolvendi, ad gravissima
ac utilissima totius arithmeticae pertinere [...] tam notum est,
ut de hac re copiose loqui superfluum foret. [...] Praetereaque
scientiae dignitas requirere videtur, ut omnia subsidia ad solutionem
problematis tam elegantis ac celebris sedulo excolantur.[1]

Carl Friedrich Gauß (1801)

ANTON FELKEL hatte [die Faktorentafel ...] bis zu zwei Millionen
in der Handschrift vollendet [...] ; allein was davon in Wien auf
öffentliche Kosten bereits gedruckt war, wurde, weil sich keine Käufer
fanden, im Türkenkriege zu Patronen verbraucht![2]

Carl Friedrich Gauß (1812)

Factoring is the resolving of a composite number into its factors,
and is performed by division.

Daniel W. Fish (1874)

L'équation $x^2 - y^2 = N$, est de la plus grande importance
dans les questions de factorisation.[3]

Maurice Kraïtchik (1926)

They made another calculation: how long it would take
to complete the job. The answer: Seven years.
They persuaded the professor to set the project aside.

Richard Phillips Feynman (1984)

[1] The problem of distinguishing prime numbers from composite numbers and of resolving the latter into their prime factors is so well known to be one of the most important and useful in arithmetic [...] that it is superfluous to speak at length about this matter. [...] Further, the dignity of the science itself seems to require that every possible means be explored for the solution of such an elegant and celebrated problem.

[2] Anton Felkel [...] had completed the manuscript of his table [of factors of integers] up to two million [...] ; the parts that had been printed in Vienna with a government grant did not sell and were, unfortunately, used to make cartridges in the war against the Turks!

[3] The equation $x^2 - y^2 = N$ is of paramount importance in the factorization problem.

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